GOLDBACH PROBLEMS FOR POLIGNAC FAMILIES: TWIN, COUSIN AND SEXY SUMMANDS

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ABSTRACT. For an even integer d, let $\mathcal{P}_d$ be the set of primes p such that $p-d$ or $p+d$ is also prime; thus $\mathcal{P}_2$, $\mathcal{P}_4$ and $\mathcal{P}_6$ are the twin, cousin and sexy primes. We study the binary problem of representing an even integer n as $p+q$ with $p \in \mathcal{P}_a$ and $q \in \mathcal{P}_b$, which for $a=b=2$ is the twin-prime Goldbach problem of Zwillinger and Dubner. By an exact, integer-only witness search we determine the exceptional sets for all six pairings of twin, cousin and sexy primes up to 10^8, and for all 120 pairs of gaps $a, b \le 30$ up to 10^7. Every even $n \ge 6$ is the sum of a twin prime and a cousin prime in this range, whereas twin+twin and cousin+cousin have 34 and 36 exceptions, the largest being 4208 and 2952. We prove a structure theorem modulo 3 that explains this contrast: when $3 \nmid a$, the representations of s, $s+a$, $s+2a$ are locked together in the exact ratio $1:2:1$, so equal-gap exceptions occur in arithmetic triples, while for $a \neq b$ the class $n \equiv 0 \pmod 3$ has two independent routes. As a corollary, the twin-prime Goldbach problem for multiples of 6 is equivalent to a Goldbach problem for the index set $\{k : 6k \pm 1 \text{ both prime}\}$. We formulate a Hardy–Littlewood type conjecture for $r_{a,b}(n)$, prove that its singular series is bounded below by an explicit absolute constant $c_0 \approx 4.1512$ (halved when $a=b$), and show that the prediction, refined by inclusion–exclusion, matches the data to within 1% near 10^7 and reproduces the number of observed exceptions. Any proof of the existence part for a single pair (a,b) would establish Polignac's conjecture for gap a or gap b.

1. INTRODUCTION

Goldbach's conjecture asserts that every even integer $n \ge 4$ is a sum of two primes. It has been verified up to $4 \cdot 10^{18}$ [10], it holds for almost all even n [8], and Chen proved that every sufficiently large even n is the sum of a prime and a number with at most two prime factors [2]; a complete proof remains out of reach. A natural strengthening restricts the summands to primes with additional arithmetic structure. In 1979 Zwillinger [15] checked every even number up to 10^6 for a representation as a sum of two primes each belonging to a twin-prime pair, and found 33 exceptions, all below 5000. Dubner [4] conjectured that 4208 is the largest such exception; the list is sequence A007534 of the OEIS, which records that there are no further exceptions below 10^{11} [9]. A recent preprint considers the intermediate problem of a prime plus a twin prime [11].

Twin primes are the first of the families singled out by Polignac's conjecture [3], which predicts infinitely many prime pairs with any prescribed even difference. This suggests the following general question.

Definition 1.1. For even $d \ge 2$ let

$$\mathcal{P}_d = \{p \text{ prime} : p-d \text{ or } p+d \text{ is prime}\}.$$

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For even $a, b \geq 2$ and even n, let $r_{a,b}(n)$ be the number of ordered pairs $(p, q) \in \mathcal{P}_a \times \mathcal{P}_b$ with $p + q = n$, and let

$$E_{a,b} = \{n \geq 4 \text{ even} : r_{a,b}(n) = 0\}, \qquad n_0(a, b) = 2 + \max E_{a,b},$$

so that $n_0(a, b)$ is the least n_0 such that every even $n \geq n_0$ is represented (if $E_{a,b}$ is finite).

We call $\mathcal{P}_2$, $\mathcal{P}_4$ and $\mathcal{P}_6$ the *twin*, *cousin* and *sexy* primes. Note that $3 \in \mathcal{P}_2 \cap \mathcal{P}_4$ (via 5 and 7) but $3 \notin \mathcal{P}_6$, and 2 lies in no $\mathcal{P}_d$. The twin-prime Goldbach problem is the case $a = b = 2$.

Results. Our contributions are as follows.

(1) *Exact computations* (Section 2). We determine $E_{a,b} \cap [4, 10^8]$ for the six pairings of twin, cousin and sexy primes and $E_{a,b} \cap [4, 10^7]$ for all pairs $2 \leq a \leq b \leq 30$, using an integer-only witness search. The thresholds are summarised in Tables 1 and 2. In particular, every even $n \geq 6$ up to 10^8 is the sum of a twin prime and a cousin prime, while the twin+twin and cousin+cousin problems have 34 and 36 exceptions.

(2) *A structure theorem modulo* 3 (Section 3). For $3 \nmid ab$ we express $r_{a,b}$, up to a bounded error, through a single convolution $N_{a,b}$ of "lower pair members" (Theorem 3.4). For $a = b$ this gives the exact identity $r^*(s) : r^*(s + a) : r^*(s + 2a) = 1 : 2 : 1$, which forces equal-gap exceptions to occur in arithmetic triples (Corollary 3.5), and it shows that the twin-prime Goldbach problem for multiples of 6 is equivalent to a Goldbach problem for the set of indices k with $6k \pm 1$ prime (Corollary 3.7). For $a \neq b$ the residue class 0 mod 3 has two independent routes (Corollary 3.6), which the data confirm.

(3) *A conjecture with explicit singular series* (Section 4). We state a Hardy–Littlewood type asymptotic for $r_{a,b}(n)$ (Conjecture 4.3) and prove that its singular series is bounded below by an absolute constant $c_0 \approx 4.1512$, halved when $a = b$ (Lemma 4.2). There is therefore no local obstruction for any pair of gaps.

(4) *Numerical evidence* (Section 5). The prediction, refined by inclusion–exclusion over the membership conditions, matches $r_{a,b}(n)$ to within 1% for n near 10^7 for all six pairings, with correlation above 0.998. A Poisson model built from it predicts 16.3 and 16.2 exceptions above 1000 for twin+twin and cousin+cousin; the observed numbers are 18 and 19.

(5) *Limits* (Section 6). The existence statement for any single pair (a, b) implies Polignac's conjecture for gap a or gap b (Proposition 3.1), so it is at least as hard as the twin-prime conjecture in the case $a = b = 2$.

2. Computations

2.1. **Method.** We sieve the primes up to $N + \max(a, b)$ and form the indicator arrays of $\mathcal{P}_a$ and $\mathcal{P}_b$ on $[0, N]$. Let U be the set of even $n \in [4, N]$, and process the elements $p \in \mathcal{P}_a$ in increasing order. At each step we remove from U every n with $n - p \in \mathcal{P}_b$, recording p as the minimal witness for n; and we declare exceptional every $n \in U$ with $n < p + \min \mathcal{P}_b$, since no later element of $\mathcal{P}_a$ can represent it. The procedure stops when U is empty. It uses integer arithmetic only, every non-exception receives an explicit witness, and every exception is certified by exhaustion. For the statistics of $r_{a,b}(n)$ in Section 5 we compute the full convolution with a fast Fourier transform in double precision and round to integers; on the overlapping range the two methods give identical exceptional sets. The exceptional set for $(2, 2)$ coincides with A007534 [9], which serves as a check.

TABLE 1. The six pairings of twin ($a = 2$), cousin ($a = 4$) and sexy ($a = 6$) primes, verified exactly to 10^8. The last column gives the largest minimal witness: the largest value, over even $n \in [n_0, 10^8]$, of the least $p \in \mathcal{P}_a$ with $n - p \in \mathcal{P}_b$.

Pairing (a, b)	$n_0(a, b)$	$\lvert E_{a,b}\rvert$	largest minimal witness p (at n)
twin + twin $(2, 2)$	4210	34	117 703 (64 752 656)
cousin + cousin $(4, 4)$	2954	36	120 623 (80 759 296)
sexy + sexy $(6, 6)$	10	3	26 833 (50 059 412)
twin + cousin $(2, 4)$	6	1	115 021 (94 270 994)
twin + sexy $(2, 6)$	8	2	33 329 (89 769 676)
cousin + sexy $(4, 6)$	8	2	40 849 (59 836 418)

2.2. **Twin, cousin and sexy pairings.**

Theorem 2.1. *For the six pairings of $\mathcal{P}_2, \mathcal{P}_4, \mathcal{P}_6$, the sets $E_{a,b} \cap [4, 10^8]$ are as follows:*

$$E_{2,4} = \{4\}, \qquad E_{2,6} = E_{4,6} = \{4, 6\}, \qquad E_{6,6} = \{4, 6, 8\},$$

$$\begin{aligned} E_{2,2} = \{&4, 94, 96, 98, 400, 402, 404, 514, 516, 518, 784, 786, 788, 904, 906, 908,\\ &1114, 1116, 1118, 1144, 1146, 1148, 1264, 1266, 1268, 1354, 1356, 1358,\\ &3244, 3246, 3248, 4204, 4206, 4208\}, \end{aligned}$$

$$\begin{aligned} E_{4,4} = \{&4, 8, 12, 68, 72, 76, 278, 282, 286, 638, 642, 644, 648, 652, 908, 912, 916,\\ &1058, 1062, 1066, 1328, 1332, 1336, 1418, 1422, 2114, 2118, 2122,\\ &2294, 2298, 2302, 2528, 2532, 2536, 2948, 2952\}. \end{aligned}$$

Proof. Exhaustive computation by the method above. □

The equal-gap exceptional sets have a visible structure: apart from a few small cases, they consist of triples $\{s, s + a, s + 2a\}$, such as $\{94, 96, 98\}$ for $a = 2$ and $\{68, 72, 76\}$ for $a = 4$. The occasional incomplete triple, such as $\{1418, 1422\}$ in $E_{4,4}$, arises because $1426 = 3 + 1423$ with $1423 \in \mathcal{P}_4$. Section 3 explains this.

2.3. **All gaps up to 30.** Table 2 gives $n_0(a, b)$ for all $2 \le a \le b \le 30$, computed exactly to 10^7. Two patterns stand out.

- If $3 \mid a$ or $3 \mid b$, then $n_0(a, b) \le 58$ (the maximum, 58, occurs for $(12, 12)$).
- If $3 \nmid ab$, the large thresholds occur on the diagonal $a = b$: $n_0(a, a)$ ranges from 666 ($a = 28$) to 5736 ($a = 16$), except for $a \in \{10, 20\}$ where $n_0 = 30$ and 40; there the factor $(\ell - 1)/(\ell - 2) = 4/3$ at $\ell = 5$ raises the density of $\mathcal{P}_a$. Off the diagonal large thresholds are rarer: the largest are 1944 $(2, 16)$, 1356 $(2, 28)$ and 1300 $(4, 26)$, while for instance $n_0(2, 4) = 6$.

3. Structure modulo 3

We begin with the elementary observation that limits what can be proved.

Proposition 3.1. *If $E_{a,b}$ is finite, then $\mathcal{P}_a$ or $\mathcal{P}_b$ is infinite; that is, there are infinitely many prime pairs with difference a or infinitely many with difference b. In particular, finiteness of $E_{2,2}$ implies the twin-prime conjecture.*

TABLE 2. Thresholds $n_0(a,b)$ for $2 \le a \le b \le 30$, verified exactly to 10^7. Bold entries exceed 500.

$a\backslash b$	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30
2	**4210**	6	8	6	6	8	6	**1944**	8	14	30	8	28	**1356**	10
4		**2954**	8	6	14	8	6	**654**	12	34	390	8	**1300**	30	14
6			10	8	8	10	8	8	16	14	30	10	8	30	12
8				**2326**	6	8	14	328	22	40	168	16	40	30	10
10					30	18	16	30	12	14	30	8	6	54	24
12						58	20	18	16	14	34	10	8	42	12
14							**1100**	140	22	40	30	16	40	30	10
16								**5736**	12	30	48	8	6	60	30
18									22	40	30	16	40	30	16
20										40	30	16	40	40	14
22											**2160**	42	30	60	36
24												18	16	30	12
26													**4390**	166	10
28														**666**	30
30															34

Proof. If $\mathcal{P}_a$ and $\mathcal{P}_b$ are both finite then so is $\mathcal{P}_a + \mathcal{P}_b$, which then cannot contain all sufficiently large even integers. Finally, $\mathcal{P}_d$ is infinite if and only if there are infinitely many primes p with $p+d$ prime, since every element of $\mathcal{P}_d$ is a member of such a pair. □

From now on fix even $a, b \ge 2$. Primes $p > 3$ lie in the classes $\pm 1 \bmod 3$, and when $3 \nmid a$ the two members of a pair $(p, p+a)$ with $p > 3$ are forced into opposite classes. This rigidity is the source of the structure below.

Definition 3.2. For even a with $3 \nmid a$ let

$$\mathcal{L}_a = \{p > 3 : p \text{ and } p + a \text{ prime}\}, \qquad \mathcal{U}_a = \mathcal{L}_a + a, \qquad \mathcal{E}_a = \mathcal{P}_a \setminus (\mathcal{L}_a \cup \mathcal{U}_a).$$

Lemma 3.3. *Let $3 \nmid a$. Then $\mathcal{L}_a \subseteq \{x \equiv a \pmod 3\}$ and $\mathcal{U}_a \subseteq \{x \equiv -a \pmod 3\}$; in particular $\mathcal{L}_a \cap \mathcal{U}_a = \varnothing$. Moreover $\mathcal{E}_a \subseteq \{3, a+3\}$.*

Proof. If $p \in \mathcal{L}_a$ then $p \not\equiv 0$ and $p + a \not\equiv 0 \pmod 3$, so $p \not\equiv -a$; since $a \not\equiv 0$, the only remaining nonzero class is $p \equiv a$. Hence $p + a \equiv 2a \equiv -a$, which proves the first two inclusions.

Now let $p \in \mathcal{P}_a \setminus (\mathcal{L}_a \cup \mathcal{U}_a)$. Since $2 + a$ is even and greater than 2, and $2 - a \le 0$, we have $2 \notin \mathcal{P}_a$, so p is odd. Suppose $p > 3$. If $p + a$ were prime then $p \in \mathcal{L}_a$, so $p - a$ is prime. If $p - a > 3$ then $p - a \in \mathcal{L}_a$ and $p \in \mathcal{U}_a$, a contradiction; and $p - a$ is odd and prime, so $p - a = 3$. Hence $p \in \{3, a+3\}$. □

Write $\mathcal{L} = \mathcal{L}_a$, $\mathcal{U} = \mathcal{U}_a$ and similarly $\mathcal{L}' = \mathcal{L}_b$, $\mathcal{U}' = \mathcal{U}_b$, and define for integers m

$$N_{a,b}(m) = \#\{(x,y) \in \mathcal{L}_a \times \mathcal{L}_b : x + y = m\},$$
$$r^*_{a,b}(n) = \#\{(p,q) \in (\mathcal{L} \cup \mathcal{U}) \times (\mathcal{L}' \cup \mathcal{U}') : p + q = n\}.$$

Theorem 3.4. *Let a, b be even with $3 \nmid ab$. Then for every even n:*

1. $0 \le r_{a,b}(n) - r^*_{a,b}(n) \le |\mathcal{E}_a| + |\mathcal{E}_b| \le 4$;
2. $r^*_{a,b}(n) = N_{a,b}(n) + N_{a,b}(n-a) + N_{a,b}(n-b) + N_{a,b}(n-a-b)$;
3. $N_{a,b}(m) = 0$ *unless* $m \equiv a + b \pmod 3$. *Consequently:*
   - *if $a \equiv b \pmod 3$: $r^*(n) = N(n)$ for $n \equiv -a$; $r^*(n) = N(n-a) + N(n-b)$ for $n \equiv 0$; $r^*(n) = N(n-a-b)$ for $n \equiv a \pmod 3$;*

- *if* $a \equiv -b \pmod 3$*:* $r^*(n) = N(n) + N(n-a-b)$ *for* $n \equiv 0$*;* $r^*(n) = N(n-a)$ *for* $n \equiv a$*;* $r^*(n) = N(n-b)$ *for* $n \equiv -a \pmod 3$.

Proof. (1) By Lemma 3.3, $\mathcal{P}_a$ is the disjoint union of $\mathcal{L} \cup \mathcal{U}$ and $\mathcal{E}_a$, and similarly for b. A representation counted by $r_{a,b}(n)$ but not by $r^*_{a,b}(n)$ has $p \in \mathcal{E}_a$ or $q \in \mathcal{E}_b$; for each such fixed p (respectively q) there is at most one partner.

(2) The four products $\mathcal{L} \times \mathcal{L}'$, $\mathcal{L} \times \mathcal{U}'$, $\mathcal{U} \times \mathcal{L}'$, $\mathcal{U} \times \mathcal{U}'$ partition the pairs counted by r^*, since $\mathcal{L} \cap \mathcal{U} = \mathcal{L}' \cap \mathcal{U}' = \varnothing$. Translating the second coordinate by $-b$, the first by $-a$, or both, maps them bijectively onto the pairs in $\mathcal{L} \times \mathcal{L}'$ with sums n, $n-b$, $n-a$, $n-a-b$ respectively.

(3) By Lemma 3.3, $x + y \equiv a + b \pmod 3$ for $(x, y) \in \mathcal{L} \times \mathcal{L}'$. If $a \equiv b$ then $a + b \equiv 2a \equiv -a$, and $n - c \equiv -a$ for $c \in \{0, a, b, a+b\}$ selects $c = 0$ when $n \equiv -a$, $c \in \{a, b\}$ when $n \equiv 0$, and $c = a + b$ when $n \equiv -a + 2a = a$. If $a \equiv -b$ then $a + b \equiv 0$, and the same bookkeeping gives the second case. □

Corollary 3.5 (Equal gaps: locked triples)**.** *Let* $3 \nmid a$ *and write* $N = N_{a,a}$, $r^* = r^*_{a,a}$. *For every* $s \equiv -a \pmod 3$,

$$r^*(s) = N(s), \qquad r^*(s+a) = 2N(s), \qquad r^*(s+2a) = N(s).$$

Hence the three numbers $s, s+a, s+2a$ *are either all represented using* $\mathcal{L}_a \cup \mathcal{U}_a$ *or none is, and*

$$E_{a,a} \subseteq \bigcup_{s \equiv -a \ (3),\ N(s)=0} \{s,\ s+a,\ s+2a\};$$

an element of such a triple escapes $E_{a,a}$ *only through a representation in which one summand lies in* $\mathcal{E}_a \subseteq \{3, a+3\}$.

Proof. Apply Theorem 3.4(2)–(3) with $b = a$: the two routes $N(n-a)$ and $N(n-b)$ for the class $n \equiv 0$ coincide. Then use part (1). □

For $a = b$ the residue class 0 mod 3 therefore receives a doubled count but no second chance: its representability is the same event as that of its two neighbours. For $a \neq b$ this degeneracy disappears.

Corollary 3.6 (Distinct gaps: two routes)**.** *Let* $a \neq b$ *with* $3 \nmid ab$. *Then an even* $n \equiv 0 \pmod 3$ *has* $r^*_{a,b}(n) = 0$ *only if two values of* $N_{a,b}$ *at different arguments vanish simultaneously:* $N(n-a) = N(n-b) = 0$ *if* $a \equiv b \pmod 3$, *and* $N(n) = N(n-a-b) = 0$ *if* $a \equiv -b \pmod 3$.

Heuristically, a vanishing value of N is a rare event and two vanishing values at different arguments are rarer still, so exceptions for $a \neq b$ should avoid the class 0 mod 3. The data agree: among all pairs $a < b$ in $\{2, 4, 8, 10, 14, 16, 20, 22, 26, 28\}$, the exceptions $n > a + b + 6$ up to 10^7 number 25, and only one of them is divisible by 3.

For the twin and cousin primes, Corollary 3.5 becomes a Goldbach problem for an explicit set of integers.

Corollary 3.7 (Index reformulation)**.** *Let* $K_2 = \{k \geq 1 : 6k-1,\ 6k+1$ *both prime*$\}$ *and* $K_4 = \{k \geq 1 : 6k+1,\ 6k+5$ *both prime*$\}$. *For every* $m \geq 2$,

$$6m \in \mathcal{P}_2 + \mathcal{P}_2 \iff m \in K_2 + K_2, \qquad 6m \in \mathcal{P}_4 + \mathcal{P}_4 \iff m - 1 \in K_4 + K_4.$$

In particular, every sufficiently large even integer is a sum of two twin primes if and only if every sufficiently large integer is a sum of two elements of K_2.

Proof. For $a = 2$: an element $p \in \mathcal{L}_2$ satisfies $p \equiv 2 \pmod 3$ and is odd, so $p = 6k - 1$ with $k \in K_2$, and conversely; thus $N_{2,2}(6m-2) = \#\{(k, j) \in K_2^2 : k + j = m\}$. By Corollary 3.5 with $s = 6m - 2 \equiv -2 \pmod 3$, $r^*_{2,2}(6m) = 2N_{2,2}(6m-2)$. The only element of $\mathcal{E}_2$ is 3 (as

$5 \in \mathcal{L}_2$), and $6m = 3 + q$ would need $q = 6m - 3$, which for $m \geq 2$ is divisible by 3 and greater than 3, hence not prime. So $r_{2,2}(6m) = r^*_{2,2}(6m)$, which proves the first equivalence.

For $a = 4$: $p \in \mathcal{L}_4$ means $p \equiv 1 \pmod 3$, so $p = 6k + 1$ with $k \in K_4$; thus $N_{4,4}(6M + 2) = \#\{(k, j) \in K_4^2 : k + j = M\}$. Taking $s = 6(m - 1) + 2 \equiv -4 \pmod 3$ gives $r^*_{4,4}(6m) = r^*_{4,4}(s + 4) = 2N_{4,4}(6(m - 1) + 2)$. Here $\mathcal{E}_4 = \{3\}$ (since $7 \in \mathcal{L}_4$), and $6m - 3$ is again not prime for $m \geq 2$.

For the last statement: if $K_2 + K_2$ contains every $m \geq m_0$, then by Corollary 3.5 each triple $\{6m - 2, 6m, 6m + 2\}$ with $m \geq m_0$ is represented, and these triples cover all even $n \geq 6m_0 - 2$. Conversely, if every even $n \geq n_1$ is represented, apply the first equivalence to $n = 6m$. □

Thus the exceptional triples of the twin-prime Goldbach problem correspond to the integers $m \notin K_2 + K_2$; the last of them is $m = 701$, giving $\{4204, 4206, 4208\}$. Similarly the triple $\{68, 72, 76\}$ of $E_{4,4}$ corresponds to $11 \notin K_4 + K_4$.

Remark 3.8. When $3 \mid a$, the members of a pair $(p, p + a)$ lie in the same class modulo 3, and both classes ± 1 occur. No analogue of Lemma 3.3 holds, and the singular series in the next section is roughly doubled at the prime 3. This is consistent with the small thresholds in the rows and columns of Table 2 indexed by multiples of 3.

4. A Hardy–Littlewood conjecture

4.1. **Heuristic.** Write $\mathbf{1}_{\mathcal{P}_a}(p) = \mathbf{1}_{\text{prime}}(p)\big(\mathbf{1}_{\text{prime}}(p+a)+\mathbf{1}_{\text{prime}}(p-a)-\mathbf{1}_{\text{prime}}(p+a)\mathbf{1}_{\text{prime}}(p-a)\big)$. Expanding $r_{a,b}(n) = \sum_p \mathbf{1}_{\mathcal{P}_a}(p)\mathbf{1}_{\mathcal{P}_b}(n-p)$ gives nine terms, each of which counts integers $x < n$ at which a fixed set of linear forms

$$x + h \ (h \in H_1), \qquad n - x + k \ (k \in H_2), \qquad 0 \in H_1 \cap H_2,$$

are simultaneously prime. The Hardy–Littlewood and Bateman–Horn heuristics [6, 1] predict that such a count is asymptotic to

$$C(H_1, H_2; n) \sum_{2 \leq x \leq n-2} \frac{1}{(\log x)^{|H_1|}(\log(n - x))^{|H_2|}}, \qquad C = \prod_{\ell} \frac{1 - \omega_\ell/\ell}{(1 - 1/\ell)^{|H_1|+|H_2|}}, \tag{1}$$

where ω_ℓ is the number of residues of $\{-h : h \in H_1\} \cup \{n + k : k \in H_2\}$ modulo ℓ. The four terms with $|H_1| = |H_2| = 2$ dominate; the others carry an extra factor $1/\log n$. When $3 \nmid a$, the term with $H_1 = \{0, a, -a\}$ vanishes identically because $x - a, x, x + a$ cover all classes modulo 3; when $3 \mid a$ it is a genuine secondary term, corresponding to prime triples $(p - a, p, p + a)$.

Definition 4.1. For even a, b and even $n > a + b$, define

$$\mathfrak{S}_{a,b}(n) = \theta_{a,b} \sum_{\varepsilon,\delta \in \{\pm 1\}} \prod_{\ell} \beta_\ell(n; \varepsilon a, \delta b), \qquad \beta_\ell = \frac{1 - \omega_\ell/\ell}{(1 - 1/\ell)^4},$$

where $\omega_\ell = \#\big(\{0,\ -\varepsilon a,\ n,\ n + \delta b\} \bmod \ell\big)$, $\theta_{a,b} = \frac{1}{2}$ if $a = b$ and $\theta_{a,b} = 1$ otherwise, and

$$I(n) = \sum_{2 \leq x \leq n-2} \frac{1}{(\log x)^2(\log(n - x))^2} \sim \frac{n}{(\log n)^4}.$$

The factor $\frac{1}{2}$ for $a = b$ converts ordered pairs to unordered ones; we use unordered counts for $a = b$ in Section 5 and Fig. 1.

Lemma 4.2. *Let a, b be even and $n > a + b$ even. Each product $\prod_\ell \beta_\ell(n; \varepsilon a, \delta b)$ converges, and*

$$\mathfrak{S}_{a,b}(n) \ \geq \ \theta_{a,b}\, c_0, \qquad c_0 = \frac{27}{2} \prod_{\ell \geq 5} \frac{1 - 4/\ell}{(1 - 1/\ell)^4} = 4.15118\ldots$$

In particular $\mathfrak{S}_{a,b}(n) > 0$: there is no local obstruction for any pair of gaps.

Proof. Fix ε, δ. The residues $0, -\varepsilon a, n, n+\delta b$ have pairwise differences $\pm\varepsilon a$, $\pm\delta b$, $\pm n$, $\pm(n+\varepsilon a)$, $\pm(n+\delta b)$, $\pm(n+\varepsilon a+\delta b)$, all nonzero because $n > a+b$. Let D be the product of these six differences. For $\ell \geq 5$ with $\ell \nmid D$ we have $\omega_\ell = 4$ and $\beta_\ell = 1 - 6/\ell^2 + O(\ell^{-3})$, so the product converges absolutely; for the finitely many $\ell \geq 5$ dividing D we have $\omega_\ell \leq 4 < \ell$, so $\beta_\ell > 0$. In all cases $\beta_\ell \geq (1-4/\ell)(1-1/\ell)^{-4}$ for $\ell \geq 5$.

At $\ell = 2$ all four residues are even, so $\omega_2 = 1$ and $\beta_2 = 8$. At $\ell = 3$ we have $\beta_3 = 0$ if $\omega_3 = 3$, and $\beta_3 \geq (1-2/3)(2/3)^{-4} = 27/16$ if $\omega_3 \leq 2$. Hence every term with $\omega_3 \leq 2$ is at least $8 \cdot \frac{27}{16} \prod_{\ell \geq 5}(1-4/\ell)(1-1/\ell)^{-4} = c_0$, and all terms are nonnegative. It remains to find signs with $\omega_3 \leq 2$, that is, with the set $\{0, -\varepsilon a, n, n+\delta b\}$ missing a class modulo 3.

If $3 \mid n$, the set is $\{0, -\varepsilon a, \delta b\}$ modulo 3. If $3 \mid a$ or $3 \mid b$ it has at most two elements; otherwise choose the signs with $-\varepsilon a \equiv \delta b$. If $3 \nmid n$, choose ε with $-\varepsilon a \in \{0, n\} \pmod 3$ (any ε if $3 \mid a$; otherwise the one with $-\varepsilon a \equiv n$), and choose δ with $n + \delta b \in \{0, n\} \pmod 3$ similarly. Then the set is contained in $\{0, n\}$.

Numerically, with primes up to 10^8 and the tail $\prod_{\ell > X}(1-6/\ell^2) \approx \exp(-6/(X \log X))$, we obtain $c_0 = 4.15118\ldots$. □

The bound is numerically sharp. Over the window $n \in [10^7 - 2 \cdot 10^4, 10^7)$, the minimum of $\mathfrak{S}_{2,4}(n)$ is 4.151 and that of $\mathfrak{S}_{2,2}(n)$ is $2.076 = c_0/2$ to four significant figures.

Conjecture 4.3. *Let $a, b \geq 2$ be even.*

(1) *The exceptional set $E_{a,b}$ is finite: every sufficiently large even integer is the sum of an element of $\mathcal{P}_a$ and an element of $\mathcal{P}_b$.*
(2) *As $n \to \infty$ through even integers,*

$$r_{a,b}(n) = \big(1 + o(1)\big)\, \mathfrak{S}_{a,b}(n)\, I(n),$$

with ordered counts for $a \neq b$ and unordered counts for $a = b$.

By Lemma 4.2, part (2) implies part (1), and in fact $r_{a,b}(n) \gg n/(\log n)^4$. For finite n we compare the data with the refined prediction $\widetilde{r}_{a,b}(n)$ obtained by summing all nine inclusion–exclusion terms (1) with signs (and halving when $a = b$). It agrees with $\mathfrak{S}_{a,b}(n)I(n)$ to leading order.

5. Numerical evidence

5.1. **Fit of the asymptotic.** We computed $r_{a,b}(n)$ for all even $n \leq 10^7$ and evaluated $\widetilde{r}_{a,b}(n)$ on five windows: $[10^3, 10^4)$, $[10^4, 10^5)$, and $[10^k, 10^k + 2\cdot 10^4)$ for $k = 5, 6$, and $[10^7 - 2\cdot 10^4, 10^7)$. The products over ℓ were taken exactly for $\ell \leq 2000$ and by their generic factors for $2000 < \ell < 10^6$. Table 3 and Fig. 1(b) show the mean ratio $r/\widetilde{r}$ on each window. In every case it approaches 1 as n grows (monotonically, apart from a dip from 0.967 to 0.965 for $(6,6)$), reaching 0.994–1.009 near 10^7, with window standard deviation 0.013–0.032 and correlation at least 0.9989 between r and $\widetilde{r}$.

The inclusion–exclusion refinement matters only when 3 divides a gap. With the four leading terms alone, the sexy pairings are overestimated by 7–13% near 10^7: the leading-term ratios are 0.867, 0.928 and 0.931 for $(6,6)$, $(2,6)$ and $(4,6)$. The correction is the secondary term from prime triples $(p-6, p, p+6)$, which is absent when $3 \nmid a$.

5.2. **Size of the singular series.** On the window $[10^7 - 2\cdot 10^4, 10^7)$, the minimum and mean of $\mathfrak{S}_{a,b}(n)$ are 2.08 and 6.97 for $(2,2)$ and $(4,4)$; 4.15 and 13.94 for $(2,4)$; and 12.45, 8.30, 10.38 and 27.88 (common mean) for $(2,6)$, $(4,6)$, $(6,6)$. The twin+cousin values are exactly twice the twin+twin values. Since the two families have the same density, this factor of two is entirely

TABLE 3. Mean of $r_{a,b}(n)/\widetilde{r}_{a,b}(n)$ over the windows described in the text (unordered counts for $a = b$). Parenthesised values use only the four leading terms; for pairings with $3 \nmid ab$ the two coincide. The last column is the correlation between r and $\widetilde{r}$ on the final window.

(a,b)	$[10^3, 10^4)$	$[10^4, 10^5)$	$\approx 10^5$	$\approx 10^6$	$\approx 10^7$	corr. at 10^7
$(2,2)$	0.889	0.939	0.967	0.990	1.009	0.9989
$(4,4)$	0.860	0.928	0.959	0.978	0.997	0.9989
$(2,4)$	0.880	0.936	0.963	0.981	1.001	0.9992
$(2,6)$	0.908 (0.734)	0.951 (0.834)	0.966 (0.860)	0.987 (0.902)	1.000 (0.928)	0.9995
$(4,6)$	0.894 (0.738)	0.948 (0.842)	0.963 (0.868)	0.982 (0.906)	0.996 (0.931)	0.9997
$(6,6)$	0.937 (0.630)	0.967 (0.759)	0.965 (0.781)	0.986 (0.836)	0.994 (0.867)	0.9995

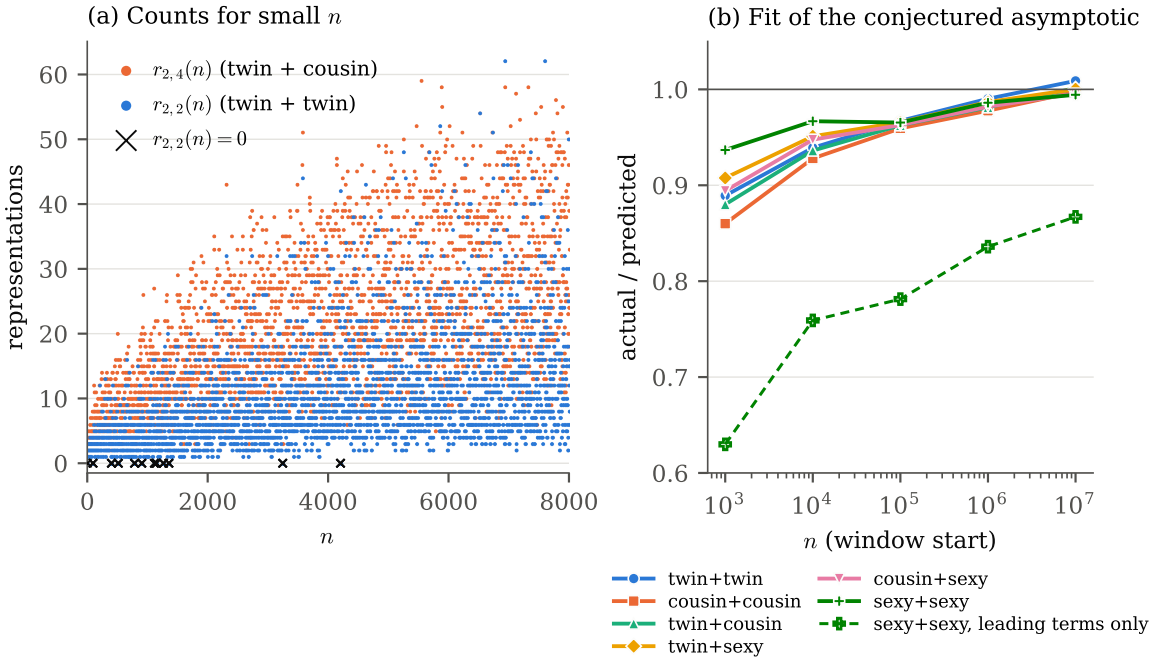


FIGURE 1. (a) Unordered counts $r_{2,2}(n)$ (twin + twin) and ordered counts $r_{2,4}(n)$ (twin + cousin) for even $n \le 8000$; crosses mark $r_{2,2}(n) = 0$. (b) Mean ratio of actual to predicted counts over the windows of Table 3. The dashed curve uses only the four leading terms for $(6,6)$.

the symmetry factor $\theta_{2,2} = \frac{1}{2}$. The theorem of Section 3 shows that this loss is not merely a matter of counting: for $a = b$ the class 0 mod 3 loses its second route.

5.3. **Number of exceptions.** Model $r_{a,b}(n)$ as Poisson with mean $\widetilde{r}_{a,b}(n)$; this is the heuristic suggested by Gallagher's analysis of prime tuples [5]. The expected number of exceptions is then $\sum_n e^{-\widetilde{r}_{a,b}(n)}$. For $n > 1000$ this gives 16.3 for $(2,2)$ and 16.2 for $(4,4)$, against 18 and 19 observed; 0.19 for $(2,4)$, against 0 observed; and less than 0.003 for each of the three sexy pairings, against 0. For $n > 10^4$ the model gives about 1.0 for $(2,2)$, while none occur below 10^{11} [9]. This is not a discrepancy. By Corollary 3.5 exceptions arrive in triples whose members

are the same event, so the expected number of exceptional *triples* beyond 10^4 is roughly half of this sum, and seeing none has probability about 0.6.

5.4. **Thresholds across gaps.** For each of the 120 pairs of Table 2 we computed $\min \mathfrak{S}_{a,b}(n)$ over $2 \cdot 30030$ consecutive even n near $2 \cdot 10^6$, a range covering every residue class modulo $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$, with the products over ℓ truncated at $\ell \le 200$. The correlation between $\log \min \mathfrak{S}_{a,b}$ and $\log n_0(a,b)$ is -0.61. The singular series explains the coarse ordering. Equal gaps with $3 \nmid a$ have the smallest minima (2.09 for $a \in \{2,4,8,16\}$, rising to 3.48 for $a \in \{14,28\}$) and the largest thresholds. For $a = b \in \{10,20\}$ the minimum is 6.26, larger than the 4.17 of twin+cousin, and the thresholds are small. Pairs with $3 \mid ab$ have minima of at least 10.4. The singular series does not, however, determine individual thresholds. The pairs $(2,4)$ and $(2,16)$ have the same minimum and mean of $\mathfrak{S}$, yet $n_0 = 6$ and $n_0 = 1944$. The Poisson model gives 0.19 expected exceptions above 1000 for $(2,4)$; with 120 pairs, a few late exceptions off the diagonal are to be expected.

6. Discussion

6.1. **On provability.** Proposition 3.1 shows that Conjecture 4.3(1) for any single pair (a,b) implies Polignac's conjecture for gap a or gap b. By the work of Zhang, Maynard and the Polymath project [14, 7, 12], some gap $d \le 246$ has infinitely many prime pairs, but no specific gap is known to. Even conditionally on strong forms of the Hardy–Littlewood conjectures, the circle method [13] handles binary problems only for almost all n, as in [8], not for all large n. Inductive arguments face a more basic obstacle. Representability is not hereditary: $4202 \in \mathcal{P}_2 + \mathcal{P}_2$ but $4204 \notin \mathcal{P}_2 + \mathcal{P}_2$. Above $n_0(a,b)$, the inductive step $n \mapsto n+2$ is equivalent to the conjecture itself.

What can be proved is structural. These are the absence of local obstructions (Lemma 4.2), the exact $1:2:1$ locking of equal-gap representations and the two-route phenomenon for distinct gaps (Theorem 3.4 and its corollaries), and the reduction of the twin-prime Goldbach problem for multiples of 6 to an additive problem for the index set K_2 (Corollary 3.7).

6.2. **Open problems.**

(1) Prove that $\{n : r_{a,b}(n) = 0\}$ has density zero, conditionally on a suitable form of the Hardy–Littlewood conjectures with uniformity in the shifts.
(2) Explain the individual thresholds $n_0(a,b)$, for example the late exception $1942 \in E_{2,16}$, beyond the Poisson model.
(3) Extend the structure theorem to moduli other than 3. Congruences $a \equiv \pm b \pmod{\ell}$ for small primes $\ell \ge 5$ may influence $n_0(a,b)$ off the diagonal; for example, $a + b = 30$ for $(2,28)$ and $(4,26)$.
(4) Study the ternary analogue, sums of three elements of $\mathcal{P}_a$. Here, too, Vinogradov's method is unavailable without knowing that $\mathcal{P}_a$ is infinite.

Reproducibility. The scripts accompanying this paper are `certify.py` (exact exceptional sets and minimal witnesses), `hl2.py` (refined prediction $\widetilde{r}_{a,b}$ and the statistics of Table 3), `verify_structure.py` (numerical check of Lemma 3.3 and Theorem 3.4 for all pairs of gaps in $\{2,4,8,10,14,16,20,22,26,28\}$ up to 10^6), `singular_series_grid.py` (the correlation across gaps in Section 5) and `fig.py` (Fig. 1). The computed data are included as JSON files.

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